CHAPTER 01

# Nomenclature

## Independent exercise solving

Original app content and actual Chromium screenshots. No textbook passages or textbook figures are embedded. Source labels and curriculum headings are not evidence of factual correctness.

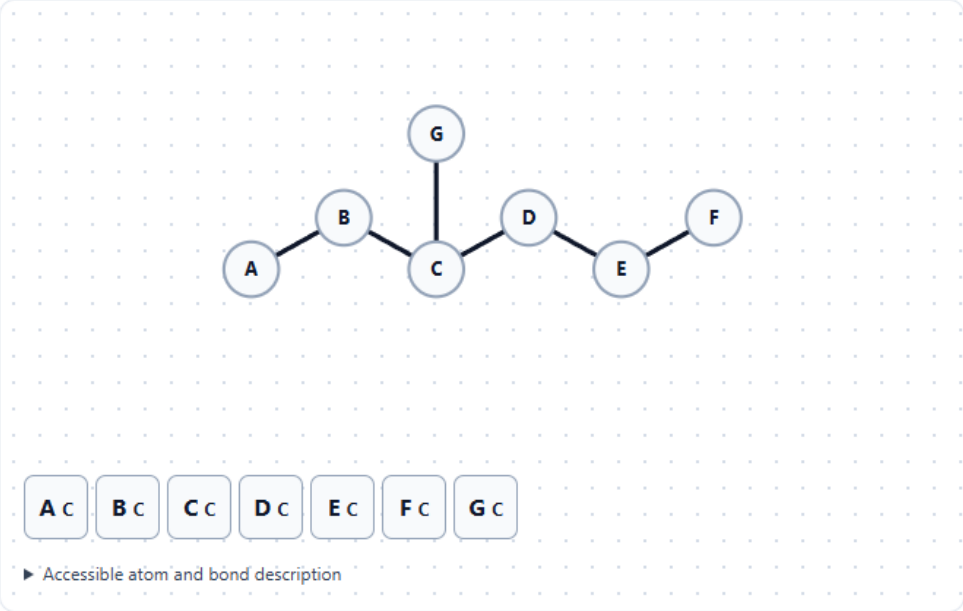
Source snapshot SHA-256: f5c68f166ab49b67d8a63fa296f0c334bb2cac4d9715811c8cb0ad7ccb9f8f40

BLIND PASS: answer keys, author molecule names/formulas, hints, explanations and answer overlays are withheld. Solve every numbered stage independently. Report the stage ID, answer, derivation, and uncertainty. Question wording and displayed diagram captions are retained because they are visible task data.

Do not open the full review packet in this chat before saving your independent attempt. The full packet contains keys and lessons that would contaminate a blind attempt.

# gp1a | Independent exercise

Screenshot: gp1a-stage-1-molecule




Graph uses zero-based indices; A=0, B=1, etc. Coordinates are drawing positions, not 3D coordinates.

0 (A): C; x=48.00, y=115.00; charge=0; explicit H override=none

1 (B): C; x=102.00, y=85.00; charge=0; explicit H override=none

2 (C): C; x=156.00, y=115.00; charge=0; explicit H override=none

3 (D): C; x=210.00, y=85.00; charge=0; explicit H override=none

4 (E): C; x=264.00, y=115.00; charge=0; explicit H override=none

5 (F): C; x=318.00, y=85.00; charge=0; explicit H override=none

6 (G): C; x=156.00, y=36.00; charge=0; explicit H override=none

Bonds [atom A, atom B, order]: [[0, 1, 1], [1, 2, 1], [2, 3, 1], [3, 4, 1], [4, 5, 1], [2, 6, 1]]

## gp1a:stage-1 | Stage 1/3: Click all 6 carbon atoms in the longest continuous parent chain.

Answer type: atoms

## gp1a:stage-2 | Stage 2/3: Which numbering direction gives the methyl branch its lowest locant?

Answer type: choice

Option 0 (A): Left-to-right (methyl at C3)

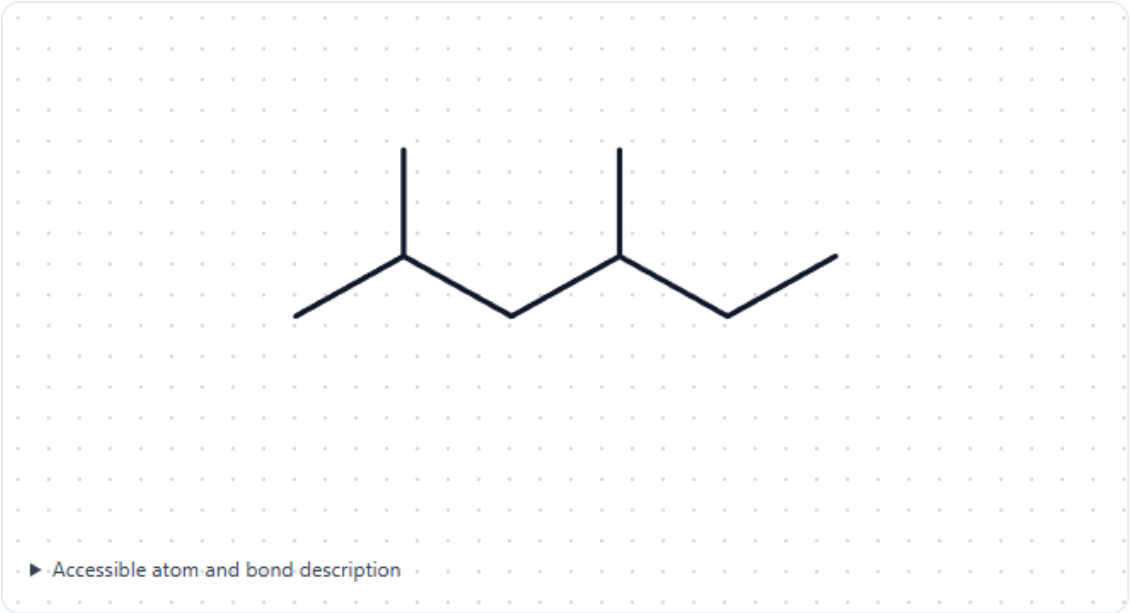
Option 1 (B): Right-to-left (methyl at C4)

## gp1a:stage-3 | Stage 3/3: Type the complete IUPAC name.

Answer type: text

# gp1b | Independent exercise

Screenshot: gp1b-stage-1-molecule




Graph uses zero-based indices; A=0, B=1, etc. Coordinates are drawing positions, not 3D coordinates.

0 (A): C; x=48.00, y=115.00; charge=0; explicit H override=none

1 (B): C; x=102.00, y=85.00; charge=0; explicit H override=none

2 (C): C; x=156.00, y=115.00; charge=0; explicit H override=none

3 (D): C; x=210.00, y=85.00; charge=0; explicit H override=none

4 (E): C; x=264.00, y=115.00; charge=0; explicit H override=none

5 (F): C; x=318.00, y=85.00; charge=0; explicit H override=none

6 (G): C; x=102.00, y=32.00; charge=0; explicit H override=none

7 (H): C; x=210.00, y=32.00; charge=0; explicit H override=none

Bonds [atom A, atom B, order]: [[0, 1, 1], [1, 2, 1], [2, 3, 1], [3, 4, 1], [4, 5, 1], [1, 6, 1], [3, 7, 1]]

## gp1b:stage-1 | Stage 1/3: Why is "2-ethyl-4-methylpentane" an invalid name for this structure?

Answer type: choice

Option 0 (A): The 2-carbon branch extends the longest continuous chain to 6 carbons (hexane)

Option 1 (B): It has 8 carbons in a straight chain

## gp1b:stage-2 | Stage 2/3: Which locant set is lower for the two methyl substituents?

Answer type: choice

Option 0 (A): Left-to-right: (2,4)

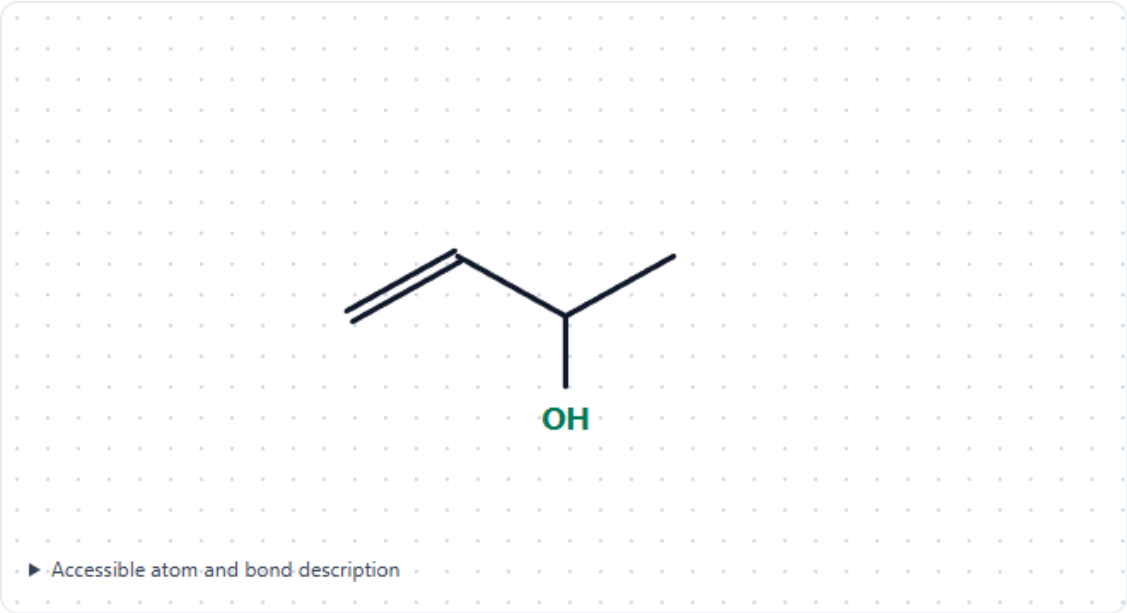
Option 1 (B): Right-to-left: (3,5)

## gp1b:stage-3 | Stage 3/3: Type the true IUPAC name.

Answer type: text

# gp2a | Independent exercise

Screenshot: gp2a-stage-1-molecule




Graph uses zero-based indices; A=0, B=1, etc. Coordinates are drawing positions, not 3D coordinates.

0 (A): C; x=48.00, y=115.00; charge=0; explicit H override=none

1 (B): C; x=102.00, y=85.00; charge=0; explicit H override=none

2 (C): C; x=156.00, y=115.00; charge=0; explicit H override=none

3 (D): C; x=210.00, y=85.00; charge=0; explicit H override=none

4 (E): O; x=156.00, y=166.00; charge=0; explicit H override=none

Bonds [atom A, atom B, order]: [[0, 1, 2], [1, 2, 1], [2, 3, 1], [2, 4, 1]]

## gp2a:stage-1 | Stage 1/3: Which group has highest priority and supplies the principal suffix?

Answer type: choice

Option 0 (A): Double bond (–ene)

Option 1 (B): Hydroxyl group (–ol; alkene kept as –en–)

## gp2a:stage-2 | Stage 2/3: Click the carbon atom on the diagram that must be assigned Carbon #1.

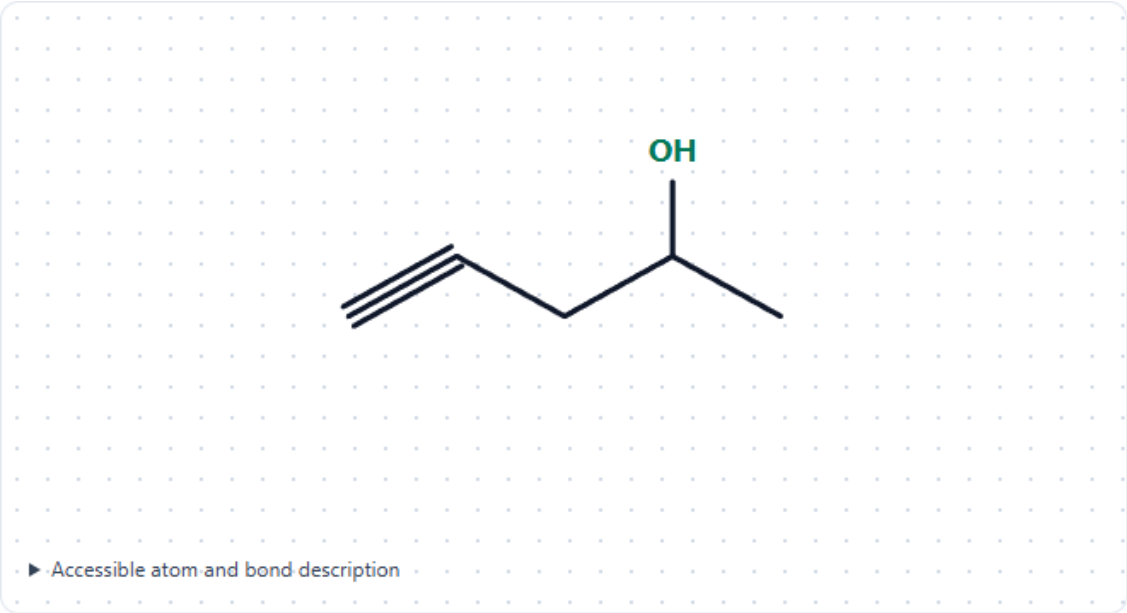
Answer type: atoms

## gp2a:stage-3 | Stage 3/3: Type the complete systematic IUPAC name.

Answer type: text

# gp2b | Independent exercise

Screenshot: gp2b-stage-1-molecule




Graph uses zero-based indices; A=0, B=1, etc. Coordinates are drawing positions, not 3D coordinates.

0 (A): C; x=48.00, y=115.00; charge=0; explicit H override=none

1 (B): C; x=102.00, y=85.00; charge=0; explicit H override=none

2 (C): C; x=156.00, y=115.00; charge=0; explicit H override=none

3 (D): C; x=210.00, y=85.00; charge=0; explicit H override=none

4 (E): C; x=264.00, y=115.00; charge=0; explicit H override=none

5 (F): O; x=210.00, y=32.00; charge=0; explicit H override=none

Bonds [atom A, atom B, order]: [[0, 1, 3], [1, 2, 1], [2, 3, 1], [3, 4, 1], [3, 5, 1]]

## gp2b:stage-1 | Stage 1/3: Which group determines the principal suffix in this 5-carbon structure?

Answer type: choice

Option 0 (A): The alcohol (–ol; triple bond becomes –yn– infix)

Option 1 (B): The alkyne (–yne; alcohol becomes hydroxy–)

## gp2b:stage-2 | Stage 2/3: Click the carbon atom that must be assigned Carbon #1.

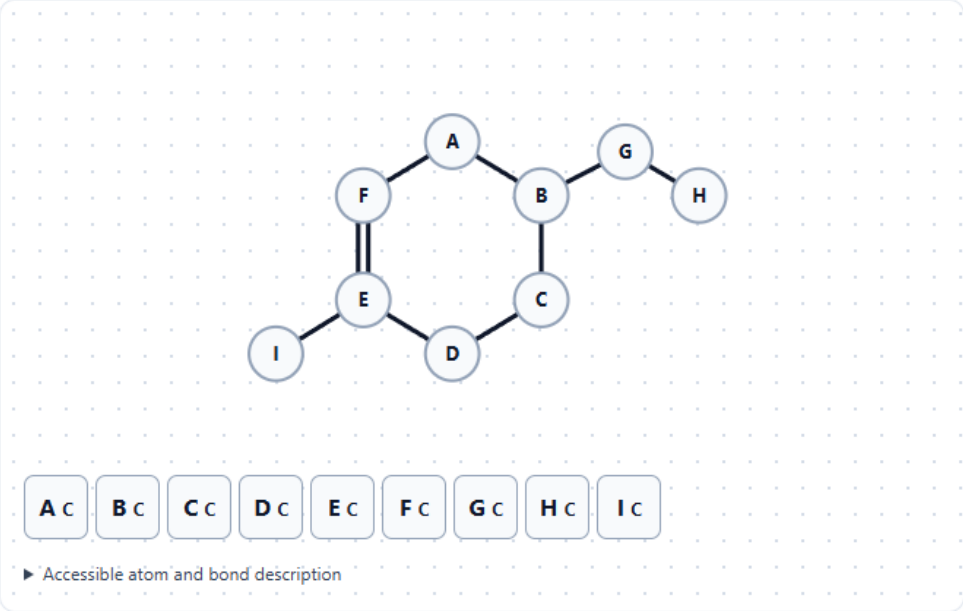
Answer type: atoms

## gp2b:stage-3 | Stage 3/3: Type the systematic IUPAC name.

Answer type: text

# gp3a | Independent exercise

Screenshot: gp3a-stage-1-molecule




Graph uses zero-based indices; A=0, B=1, etc. Coordinates are drawing positions, not 3D coordinates.

0 (A): C; x=155.00, y=42.00; charge=0; explicit H override=none

1 (B): C; x=208.00, y=74.00; charge=0; explicit H override=none

2 (C): C; x=208.00, y=136.00; charge=0; explicit H override=none

3 (D): C; x=155.00, y=168.00; charge=0; explicit H override=none

4 (E): C; x=102.00, y=136.00; charge=0; explicit H override=none

5 (F): C; x=102.00, y=74.00; charge=0; explicit H override=none

6 (G): C; x=258.00, y=48.00; charge=0; explicit H override=none

7 (H): C; x=302.00, y=74.00; charge=0; explicit H override=none

8 (I): C; x=50.00, y=168.00; charge=0; explicit H override=none

Bonds [atom A, atom B, order]: [[0, 1, 1], [1, 2, 1], [2, 3, 1], [3, 4, 1], [4, 5, 2], [5, 0, 1], [1, 6, 1], [6, 7, 1], [4, 8, 1]]

## gp3a:stage-1 | Stage 1/3: Click the ring carbon that must be assigned Carbon #1.

Answer type: atoms

## gp3a:stage-2 | Stage 2/3: With C1 at the methylated alkene carbon and C2 at the other alkene carbon, what is the ethyl locant?

Answer type: choice

Option 0 (A): C4 (giving 4-ethyl-1-methyl)

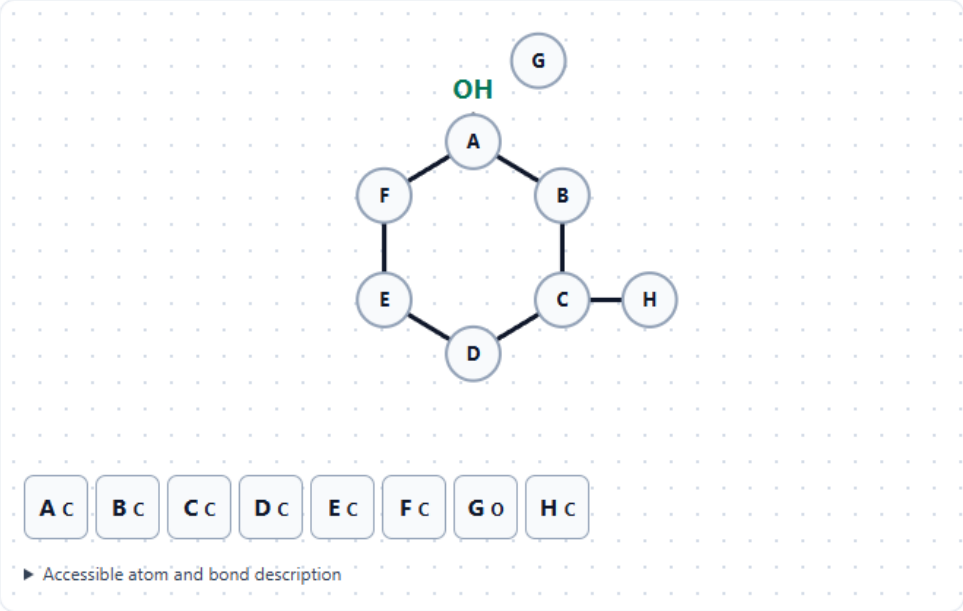
Option 1 (B): C5 (giving 5-ethyl-2-methyl)

## gp3a:stage-3 | Stage 3/3: Type the complete IUPAC name.

Answer type: text

# gp3b | Independent exercise

Screenshot: gp3b-stage-1-molecule




Graph uses zero-based indices; A=0, B=1, etc. Coordinates are drawing positions, not 3D coordinates.

0 (A): C; x=155.00, y=42.00; charge=0; explicit H override=none

1 (B): C; x=208.00, y=74.00; charge=0; explicit H override=none

2 (C): C; x=208.00, y=136.00; charge=0; explicit H override=none

3 (D): C; x=155.00, y=168.00; charge=0; explicit H override=none

4 (E): C; x=102.00, y=136.00; charge=0; explicit H override=none

5 (F): C; x=102.00, y=74.00; charge=0; explicit H override=none

6 (G): O; x=155.00, y=10.00; charge=0; explicit H override=none

7 (H): C; x=260.00, y=136.00; charge=0; explicit H override=none

Bonds [atom A, atom B, order]: [[0, 1, 1], [1, 2, 1], [2, 3, 1], [3, 4, 1], [4, 5, 1], [5, 0, 1], [0, 6, 1], [2, 7, 1]]

## gp3b:stage-1 | Stage 1/3: Click the ring carbon that must be assigned Carbon #1.

Answer type: atoms

## gp3b:stage-2 | Stage 2/3: From C1 (at –OH), which direction gives the methyl group its lowest locant?

Answer type: choice

Option 0 (A): Clockwise (methyl at C3)

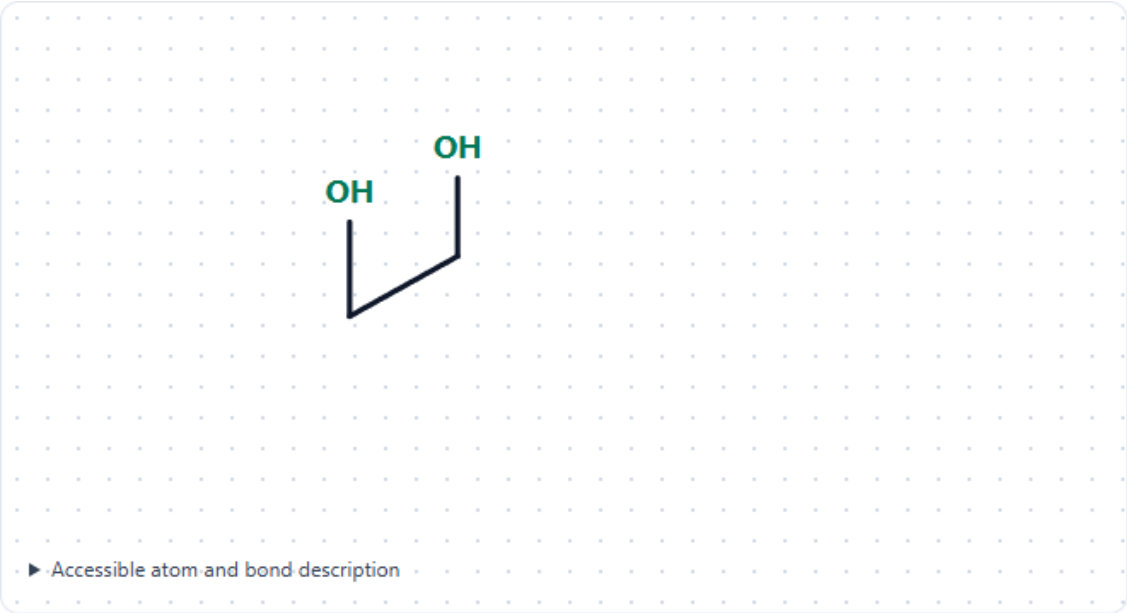
Option 1 (B): Counter-clockwise (methyl at C5)

## gp3b:stage-3 | Stage 3/3: Type the systematic IUPAC name.

Answer type: text

# gp4a | Independent exercise

Screenshot: gp4a-stage-1-molecule




Graph uses zero-based indices; A=0, B=1, etc. Coordinates are drawing positions, not 3D coordinates.

0 (A): C; x=48.00, y=115.00; charge=0; explicit H override=none

1 (B): C; x=102.00, y=85.00; charge=0; explicit H override=none

2 (C): O; x=48.00, y=52.00; charge=0; explicit H override=none

3 (D): O; x=102.00, y=30.00; charge=0; explicit H override=none

Bonds [atom A, atom B, order]: [[0, 1, 1], [0, 2, 1], [1, 3, 1]]

## gp4a:stage-1 | Stage 1/2: How are the two hydroxyl groups positioned in this molecule?

Answer type: choice

Option 0 (A): Vicinal (on adjacent carbons C1 and C2)

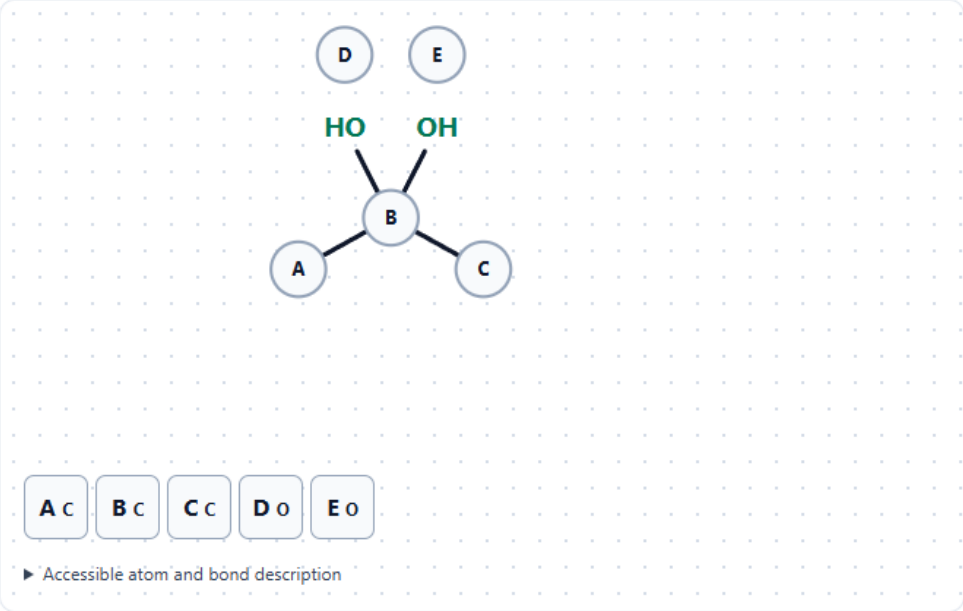
Option 1 (B): Geminal (paired on the same carbon)

## gp4a:stage-2 | Stage 2/2: Type the systematic IUPAC name (common name: ethylene glycol).

Answer type: text

# gp4b | Independent exercise

Screenshot: gp4b-stage-1-molecule




Graph uses zero-based indices; A=0, B=1, etc. Coordinates are drawing positions, not 3D coordinates.

0 (A): C; x=48.00, y=115.00; charge=0; explicit H override=none

1 (B): C; x=102.00, y=85.00; charge=0; explicit H override=none

2 (C): C; x=156.00, y=115.00; charge=0; explicit H override=none

3 (D): O; x=75.00, y=32.00; charge=0; explicit H override=none

4 (E): O; x=129.00, y=32.00; charge=0; explicit H override=none

Bonds [atom A, atom B, order]: [[0, 1, 1], [1, 2, 1], [1, 3, 1], [1, 4, 1]]

## gp4b:stage-1 | Stage 1/2: Click the carbon atom bearing both hydroxyl groups in this geminal diol.

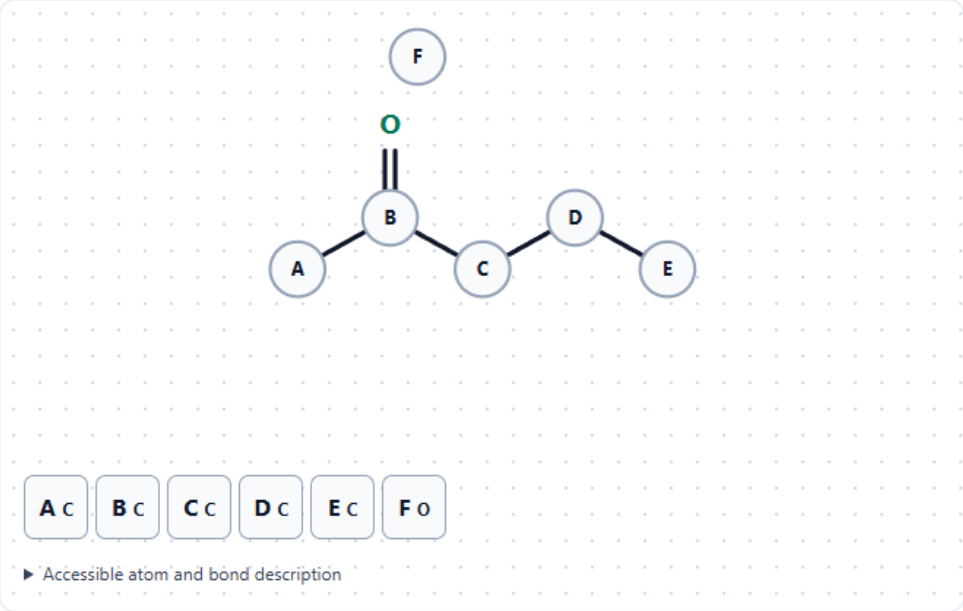
Answer type: atoms

## gp4b:stage-2 | Stage 2/2: Type the systematic IUPAC name (remember: every group needs a locant!).

Answer type: text

# gp5a | Independent exercise

Screenshot: gp5a-stage-1-molecule




Graph uses zero-based indices; A=0, B=1, etc. Coordinates are drawing positions, not 3D coordinates.

0 (A): C; x=48.00, y=115.00; charge=0; explicit H override=none

1 (B): C; x=102.00, y=85.00; charge=0; explicit H override=none

2 (C): C; x=156.00, y=115.00; charge=0; explicit H override=none

3 (D): C; x=210.00, y=85.00; charge=0; explicit H override=none

4 (E): C; x=264.00, y=115.00; charge=0; explicit H override=none

5 (F): O; x=102.00, y=30.00; charge=0; explicit H override=none

Bonds [atom A, atom B, order]: [[0, 1, 1], [1, 2, 1], [2, 3, 1], [3, 4, 1], [1, 5, 2]]

## gp5a:stage-1 | Stage 1/3: Click BOTH α-carbons on the diagram (do not click the C=O carbon).

Answer type: atoms

## gp5a:stage-2 | Stage 2/3: Which direction gives the ketone carbonyl its lower locant?

Answer type: choice

Option 0 (A): Left-to-right (C2)

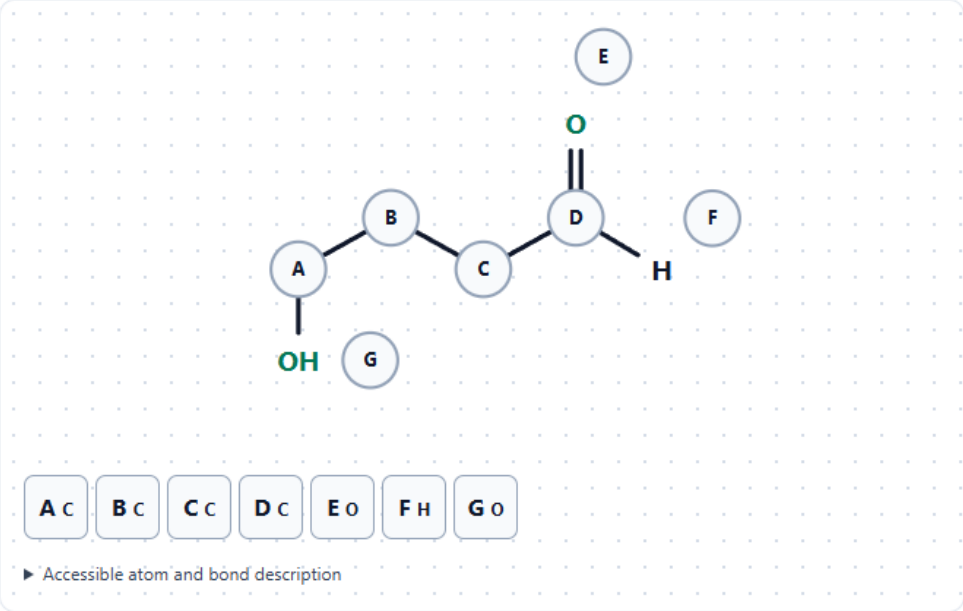
Option 1 (B): Right-to-left (C4)

## gp5a:stage-3 | Stage 3/3: Type the systematic IUPAC name.

Answer type: text

# gp5b | Independent exercise

Screenshot: gp5b-stage-1-molecule




Graph uses zero-based indices; A=0, B=1, etc. Coordinates are drawing positions, not 3D coordinates.

0 (A): C; x=48.00, y=115.00; charge=0; explicit H override=none

1 (B): C; x=102.00, y=85.00; charge=0; explicit H override=none

2 (C): C; x=156.00, y=115.00; charge=0; explicit H override=none

3 (D): C; x=210.00, y=85.00; charge=0; explicit H override=none

4 (E): O; x=210.00, y=30.00; charge=0; explicit H override=none

5 (F): H; x=260.00, y=115.00; charge=0; explicit H override=none

6 (G): O; x=48.00, y=168.00; charge=0; explicit H override=none

Bonds [atom A, atom B, order]: [[0, 1, 1], [1, 2, 1], [2, 3, 1], [3, 4, 2], [3, 5, 1], [0, 6, 1]]

## gp5b:stage-1 | Stage 1/3: Click the carbon atom that must be assigned Carbon #1.

Answer type: atoms

## gp5b:stage-2 | Stage 2/3: When an alcohol is subordinate to an aldehyde, how is the C4 –OH named?

Answer type: choice

Option 0 (A): 4-hydroxy–

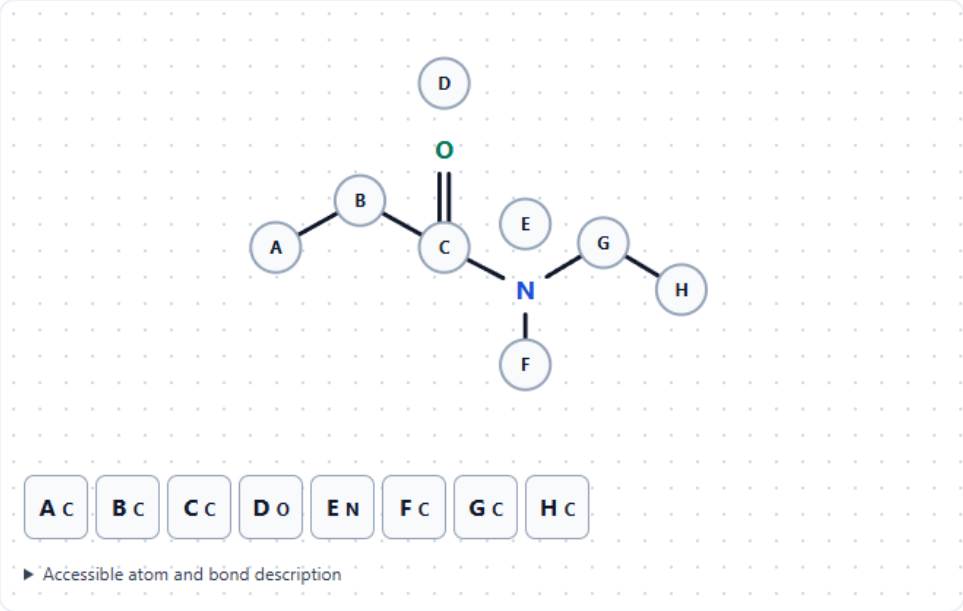
Option 1 (B): 4-oxo–

## gp5b:stage-3 | Stage 3/3: Type the complete IUPAC name.

Answer type: text

# gp6a | Independent exercise

Screenshot: gp6a-stage-1-molecule




Graph uses zero-based indices; A=0, B=1, etc. Coordinates are drawing positions, not 3D coordinates.

0 (A): C; x=48.00, y=115.00; charge=0; explicit H override=none

1 (B): C; x=102.00, y=85.00; charge=0; explicit H override=none

2 (C): C; x=156.00, y=115.00; charge=0; explicit H override=none

3 (D): O; x=156.00, y=52.00; charge=0; explicit H override=none

4 (E): N; x=208.00, y=142.00; charge=0; explicit H override=none

5 (F): C; x=208.00, y=190.00; charge=0; explicit H override=none

6 (G): C; x=258.00, y=112.00; charge=0; explicit H override=none

7 (H): C; x=308.00, y=142.00; charge=0; explicit H override=none

Bonds [atom A, atom B, order]: [[0, 1, 1], [1, 2, 1], [2, 3, 2], [2, 4, 1], [4, 5, 1], [4, 6, 1], [6, 7, 1]]

## gp6a:stage-1 | Stage 1/3: Click the carbonyl carbon (Carbon #1 of the amide parent chain).

Answer type: atoms

## gp6a:stage-2 | Stage 2/3: How are the ethyl and methyl groups on nitrogen written?

Answer type: choice

Option 0 (A): N-ethyl-N-methyl- (alphabetical)

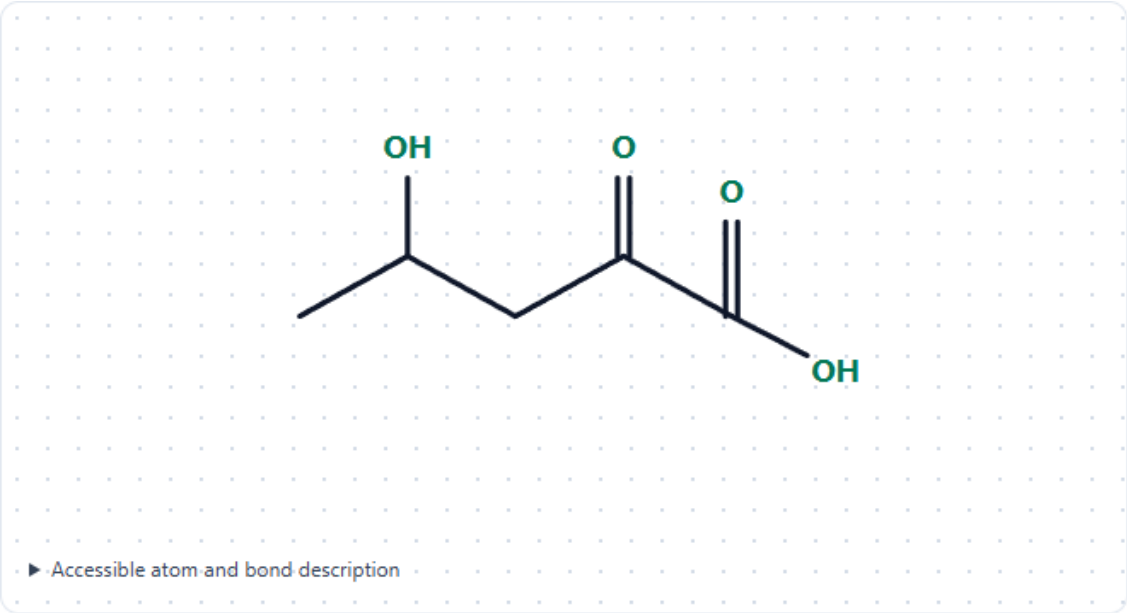
Option 1 (B): N-methyl-N-ethyl-

## gp6a:stage-3 | Stage 3/3: Type the complete IUPAC name.

Answer type: text

# gp6b | Independent exercise

Screenshot: gp6b-stage-1-molecule




Graph uses zero-based indices; A=0, B=1, etc. Coordinates are drawing positions, not 3D coordinates.

0 (A): C; x=48.00, y=115.00; charge=0; explicit H override=none

1 (B): C; x=102.00, y=85.00; charge=0; explicit H override=none

2 (C): C; x=156.00, y=115.00; charge=0; explicit H override=none

3 (D): C; x=210.00, y=85.00; charge=0; explicit H override=none

4 (E): C; x=264.00, y=115.00; charge=0; explicit H override=none

5 (F): O; x=102.00, y=30.00; charge=0; explicit H override=none

6 (G): O; x=210.00, y=30.00; charge=0; explicit H override=none

7 (H): O; x=264.00, y=52.00; charge=0; explicit H override=none

8 (I): O; x=316.00, y=142.00; charge=0; explicit H override=none

Bonds [atom A, atom B, order]: [[0, 1, 1], [1, 2, 1], [2, 3, 1], [3, 4, 1], [1, 5, 1], [3, 6, 2], [4, 7, 2], [4, 8, 1]]

## gp6b:stage-1 | Stage 1/3: Which group has highest priority and determines the suffix?

Answer type: choice

Option 0 (A): Carboxylic acid (–oic acid)

Option 1 (B): Ketone (–one)

## gp6b:stage-2 | Stage 2/3: With C1 at the acid, what are the prefixes for the C2 ketone and C4 alcohol?

Answer type: choice

Option 0 (A): 2-oxo– and 4-hydroxy–

Option 1 (B): 4-oxo– and 2-hydroxy–

## gp6b:stage-3 | Stage 3/3: Type the complete IUPAC name.

Answer type: text